

API SECURITY RISK ANALYSIS REPORT

Future Interns – Cyber Security Internship

Submitted By

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CIN ID

FIT/JUN26/CS9221

Internship Domain

Cyber Security

Task Number

Task 3

Organization

Future Interns

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1. INTRODUCTION

Application Programming Interfaces (APIs) are widely used to enable communication between software applications. APIs exchange large amounts of data and are commonly used in web applications, mobile applications, and cloud services.

Improperly secured APIs may expose sensitive information and create security risks. This report analyzes common API security risks and provides recommendations for improving API security.

2. OBJECTIVE

The objectives of this project are:

- Understand API security concepts.
 - Identify common API security risks.
 - Analyze potential vulnerabilities.
 - Assess security impact.
 - Provide remediation recommendations.
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3. SCOPE

Target API:

JSONPlaceholder Public API

API URL:

<https://jsonplaceholder.typicode.com>

Assessment Type:

Passive API Security Review

Scope Includes:

- Data Exposure Review
 - Authentication Analysis
 - Authorization Review
 - Input Validation Review
 - General Security Assessment
-

4. METHODOLOGY

The assessment was performed using the following approach:

1. API endpoint review.
2. Browser-based API testing.
3. Data exposure analysis.
4. Security risk identification.
5. Documentation of findings and recommendations.

Tools Used:

- Browser
 - Browser Developer Tools
 - Google Search
 - GitHub
 - Microsoft Word
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5. API SECURITY FINDINGS

Finding 1

Title: Lack of Authentication

Risk Level: Medium

Description:

The public API allows users to access information without authentication. While suitable for testing purposes, production APIs should implement authentication controls.

Impact:

Unauthorized users may access API resources.

Recommendation:

Implement API Keys, OAuth, or Token-Based Authentication.

{JSON} Placeholder

Free fake and reliable API for testing and prototyping.

Powered by [JSON Server](#) + [LowDB](#).

Serving ~3 billion requests each month.

Get started

Finding 2

Title: Data Exposure

Risk Level: Low

Description:

The API returns structured data that is publicly accessible. Sensitive information should never be exposed through API responses.

Impact:

Information disclosure risks may occur if sensitive data is included.

Recommendation:

Limit API responses to necessary data only.

```
{
  "userId": 1,
  "id": 1,
  "title": "sunt aut facere repellat provident occaecati excepturi optio reprehenderit",
  "body": "quia et suscipit\nsuscipit recusandae consequuntur expedita et cum\nreprehenderit molestiae ut ut quas totam\nnostrum rerum est autem sunt rem eveniet architecto"
},
{
  "userId": 1,
  "id": 2,
  "title": "qui est esse",
  "body": "est rerum tempore vitae\nsequi sint nihil reprehenderit dolor beatae ea dolores neque\nfugiat blanditiis voluptate porro vel nihil molestiae ut reiciendis\nqui aperiam non debitis possimus qui neque  
isi nulla"
},
{
  "userId": 1,
  "id": 3,
  "title": "ea molestias quasi exercitationem repellat qui ipsa sit aut",
  "body": "et iusto sed quo iure\nvoluptatem occaecati omnis eligendi aut ad\nvoluptatem doloribus vel accusantium quis pariatur\nmolestiae porro eius odio et labore et velit aut"
},
{
  "userId": 1,
  "id": 4,
  "title": "eum et est occaecati",
  "body": "ullam et saepe reiciendis voluptatem adipisci\nsit amet autem assumenda provident rerum culpa\nquis hic commodi nesciunt rem tenetur doloremque ipsam iure\nquis sunt voluptatem rerum illo velit"
},
{
  "userId": 1,
  "id": 5,
  "title": "nesciunt quas odio",
  "body": "repudiandae veniam quaerat sunt sed\nalias aut fugiat sit autem sed est\nvoluptatem omnis possimus esse voluptatibus quis\nnest aut tenetur dolor neque"
},
{
  "userId": 1,
  "id": 6,
  "title": "dolorem eum magni eos aperiam quia",
  "body": "ut aspernatur corporis harum nihil quis provident sequi\nmollitia nobis aliquid molestiae\nperspiciatis et ea nemo ab reprehenderit accusantium quas\nvoluptate dolores velit et doloremque molestiae"
},
{
  "userId": 1,
  "id": 7,
  "title": "magnam facilis autem",
  "body": "dolore placeat quibusdam ea quo vitae\nmagni quis enim qui quis quo nemo aut saepe\nquidem repellat excepturi ut quia\nsunt ut sequi eos ea sed quas"
},
}
```

Finding 3

Title: Missing Rate Limiting Controls

Risk Level: Medium

Description:

APIs without rate limiting may be vulnerable to abuse and excessive requests.

Impact:

Attackers may overload services using automated requests.

Recommendation:

Implement request throttling and rate-limiting mechanisms.

⌕ ↺ Elements Console Sources Network 4 2 1 ↻ ⚙ ⋮ ✕

<!DOCTYPE html>
<html itemscope itemtype="http://schema.org/SearchResultsPage" lang="en-IN">
 view-source scroll
 <head>⋮</head>
 <div style="visibility: hidden;" tabindex="0"></div>
 <div id="BubbleLayer-0DE9E47C-871A-4F90-8440-B190C216800A" tabindex="0" style="display: none;">⋮</div>
 <body jsmodel="hspDDf JfINDf" class="srp" jscontroller="Eox39d" marginheight="3" topmargin="3" jsaction="rcuQ6b:npT2md;xjhTIf:.CLIENT;O2vyse:.CLIENT;IVKTfe:.CLIENT;Ez7VMc:.CLIENT;R6Slyc:.CLIENT;zbW2Cf:.CLIENT;OZ3M7e:.CLIENT;gZgazf:.CLIENT;ydzCDF:.CLIENT" id="gsr"> == \$0
 <style>⋮</style>
 <style>⋮</style>
 <script nonce>⋮</script>
 <noscript>⋮</noscript>
 <div jscontroller="EufiNb" class="wYq63b" data-ved="0ahUKEwj52ZuJ-pCVAXVzTmwGHZ_DA7EQ7LoDCAQ">⋮</div> flex
 <div class="xrOgrb"></div>
 ⋮
 </body>
html body#gsr.srp

Styles Computed Layout Event Listeners DOM Breakpoints Properties >>
Filter (*) :hov .cls + ⌵ ⌶ ⌷
element.style {
}
@media (max-width: 1124px) {
 .srp {
 --center-abs-margin: 28px;
 }
}
.srp {
 --center-abs-margin: 180px;
 --center-width: 652px;
 --rhs-margin: 76px;
 --rhs-width: 372px;
 --lhs-refinements-width: 0px;
 --lhs-margin: 24px;
 --lhs-width: calc(var(--center-abs-margin) - var(--lhs-margin) - var(--center-margin));
 --center-margin: 24px;
}

6. RISK ASSESSMENT

Security Risk	Risk Level
Lack of Authentication	Medium
Data Exposure	Low
Missing Rate Limiting	Medium

Overall Risk Rating:

Medium

The identified risks highlight the importance of implementing strong API security controls.

7. RECOMMENDATIONS

The following recommendations are provided:

- Implement secure authentication mechanisms.
 - Enforce proper authorization controls.
 - Restrict unnecessary data exposure.
 - Implement API rate limiting.
 - Validate all user inputs.
 - Monitor API traffic and logs.
 - Conduct regular API security assessments.
-

8. CONCLUSION

The API Security Risk Analysis identified common security concerns that may affect API-based applications. While the analyzed API is intended for testing and learning purposes, the assessment demonstrates important API security concepts including authentication, data exposure, and rate limiting.

Organizations should implement strong security controls to protect APIs from unauthorized access and misuse.

9. REFERENCES

1. OWASP API Security Top 10
2. OWASP Security Guidelines
3. API Security Best Practices
4. JSON Placeholder Documentation
5. Cyber Security Awareness Resources